

Who Answers the Survey?

Reweighting a voluntary response sample to match the population

100,000

app users emailed an
in-app spending survey

1,364

self-selected respondents
(1.4% response rate)

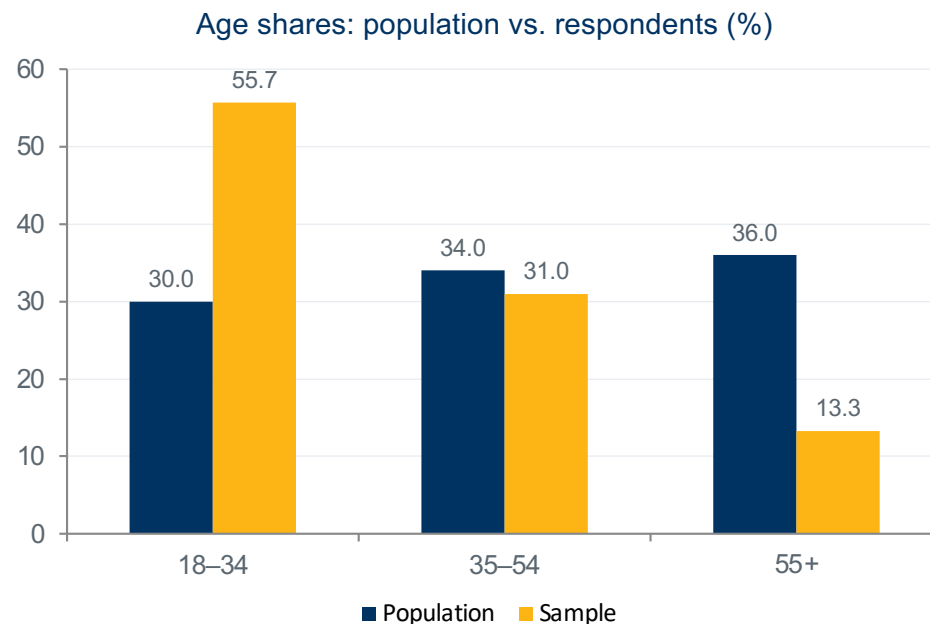
3 observables

gender × age group × education
known for sample AND population

Question of interest: What does the average user spend per month?

The Sample Doesn't Look Like the Population

Young, college-educated users answered; older users didn't



Mean monthly spend

\$92.25 unweighted sample

Why: response is voluntary, and the people most likely to respond (young, college-educated, heavy users) also spend the most. Selection into the sample is correlated with the outcome — leading to bias.

The Fix: Post-Stratification Weights

No model, no regression — one division per cell

1. Partition both sample and population into cells defined by the observables: gender × age × education ⇒ 12 cells

$$w_c = \text{population share of cell } c \div \text{sample share of cell } c$$

2. Probabilistic reading: $w_c \propto 1 / \text{Pr}(\text{respond} \mid \text{cell } c)$. If your group responded at one-fifth the average rate, you count five times over.
3. By construction, weighted cell shares match the population table exactly:

$$\bar{x}_w = \sum_i w_i x_i / \sum_i w_i \quad (\text{each respondent inherits their cell's weight})$$

Weight Construction, Cell by Cell

Underrepresented cells get weights > 1; overrepresented cells get weights < 1

Gender	Age	Education	Pop. share (%)	Sample share (%)	Weight
Female	18–34	College	5.81	14.81	0.39
Male	18–34	College	5.59	11.95	0.47
Female	18–34	No college	9.49	15.91	0.60
Male	18–34	No college	9.11	13.05	0.70
Female	35–54	College	6.94	9.38	0.74
Male	35–54	College	6.66	7.26	0.92
Male	35–54	No college	10.00	7.11	1.41
Female	35–54	No college	10.40	7.26	1.43
Female	55+	College	5.51	3.15	1.75
Male	55+	College	5.29	2.71	1.95
Male	55+	No college	12.35	3.74	3.30
Female	55+	No college	12.85	3.67	3.51

Read one row:

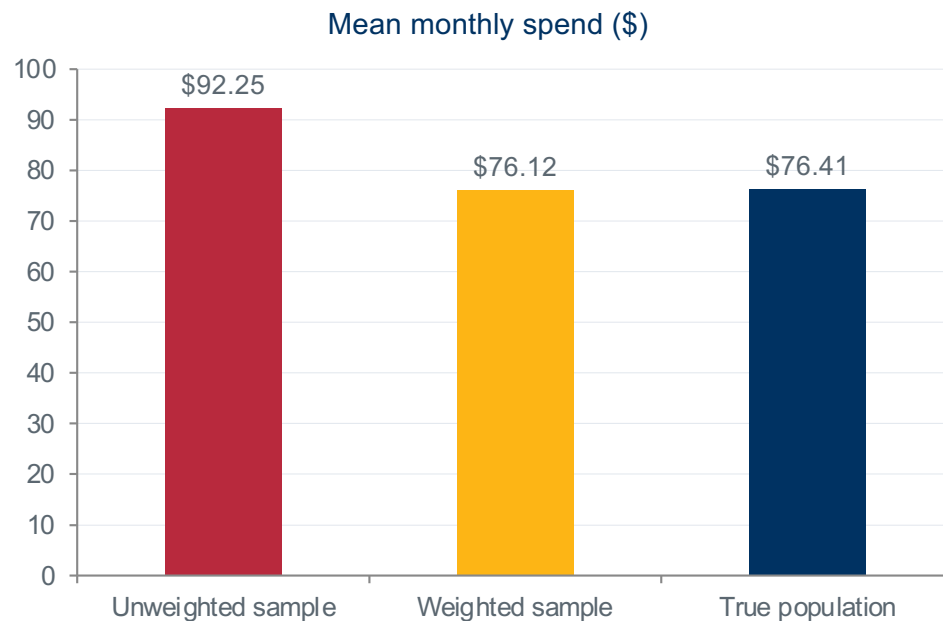
Women 55+ without a degree are 12.85% of the population but only 3.67% of respondents.

$$12.85 \div 3.67 = 3.51$$

Each such respondent stands in for 3½ “average” respondents. Weights average to 1 across the sample.

Weighted Moments Recover the Population

...along the observables we matched — and only those



It works on any outcome: share who would recommend the app falls from 66.1% (unweighted) to 60.6% (weighted).

The fine print

Weighting fixes selection on observables only. If heavy spenders respond more within every cell, the weighted mean is still biased.

Note: we know “True population” here because this is a made up / simulated example. Most often, we would not.